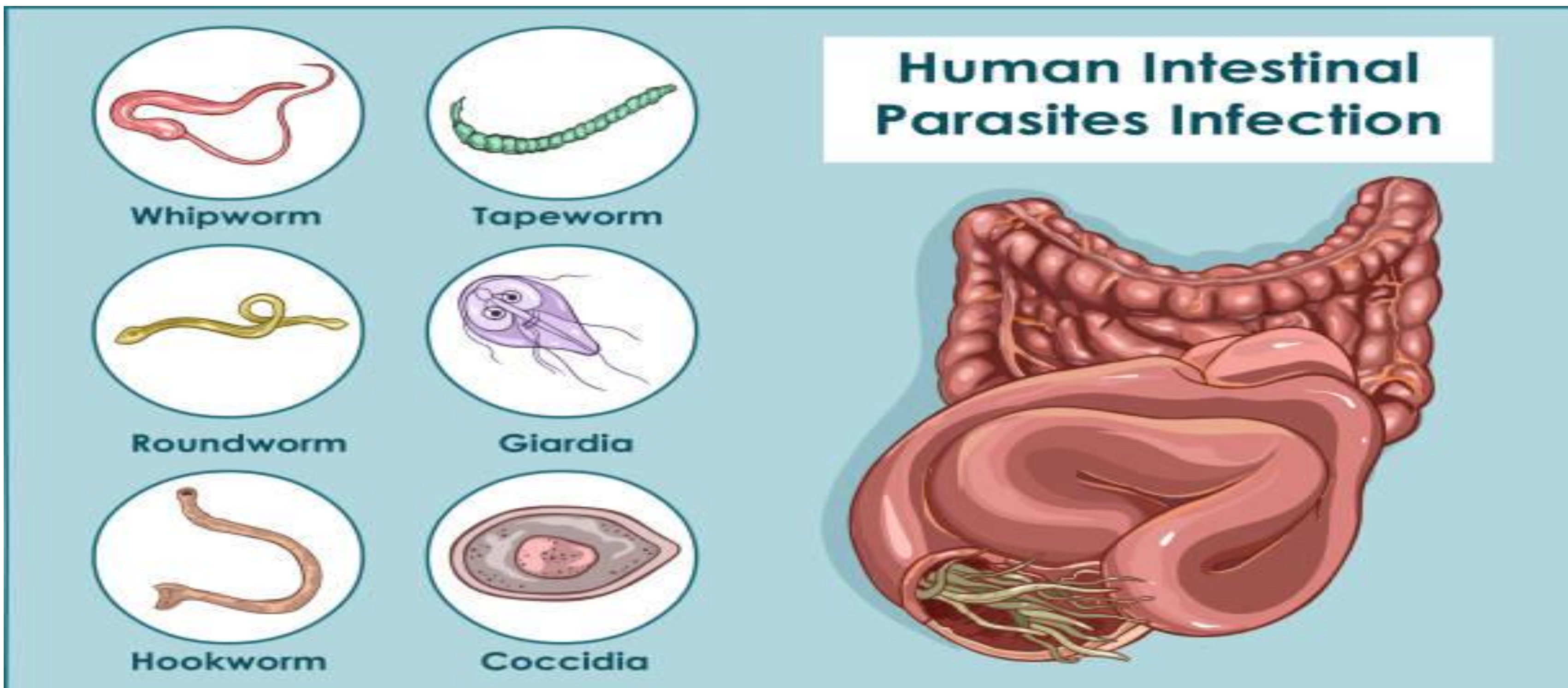
Treatment of Gastro-intestinal parasitic infestations

Dr. Begard O. Berzinji PhD Pharmacology

Parasitic infections, caused by protozoan parasites and intestinal helminths, are among the most prevalent infections in humans in developing countries. In developed countries, protozoan parasites more commonly cause gastrointestinal infections compared to helminths.

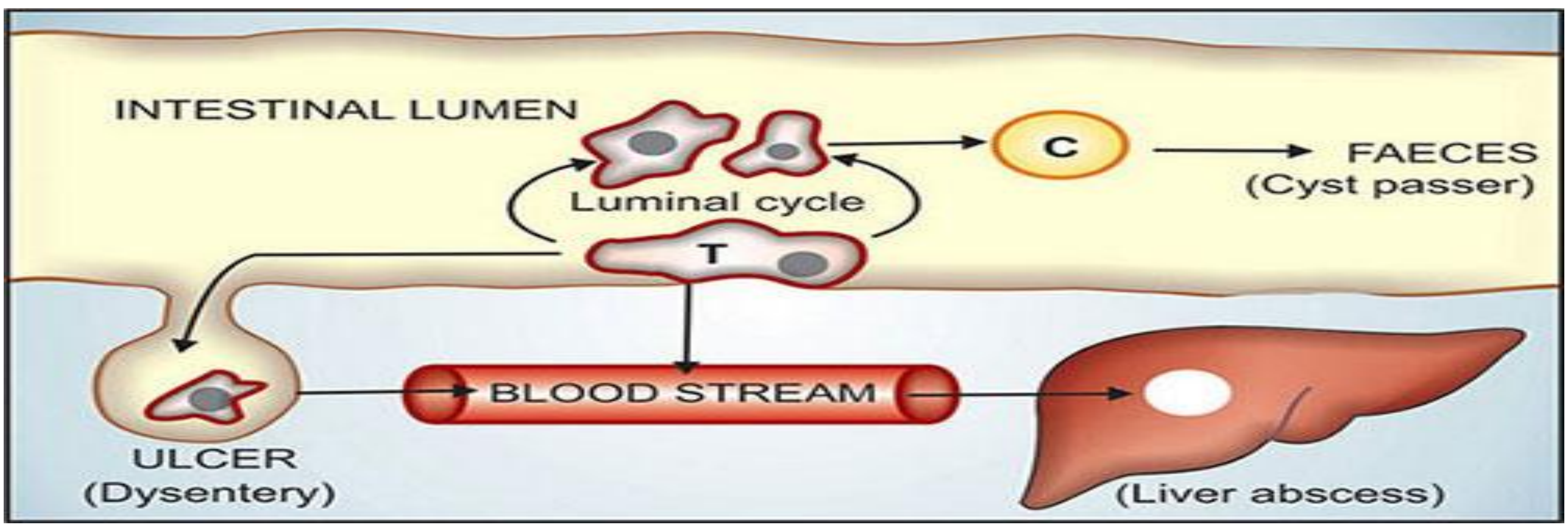
It is most commonly seen among individuals living in poverty, crowded conditions, and areas with poor sanitation.



Intestinal protozoan infections

Amebiasis *Entamoeba histolytica* This organism can cause:

- **Asymptomatic intestinal infection** but excrete the infectious cyst form, making them a source for further infections.
- **Mild to moderate or severe colitis,** the *E. histolytica* trophozoites invade into the colonic mucosa with resulting colitis and bloody diarrhea (amebic dysentery).
- **Amebic liver abscess or other extraintestinal infections**. trophozoites invade through the colonic mucosa reach the portal circulation, and travel to the liver



Metronidazole and Tinidazole:

The cornerstone of therapy for amebiasis is metronidazole or its analogue tinidazole. Metronidazole kills trophozoites but not cysts. It is clinically effective in amebiasis, trichomoniasis, and giardiasis.

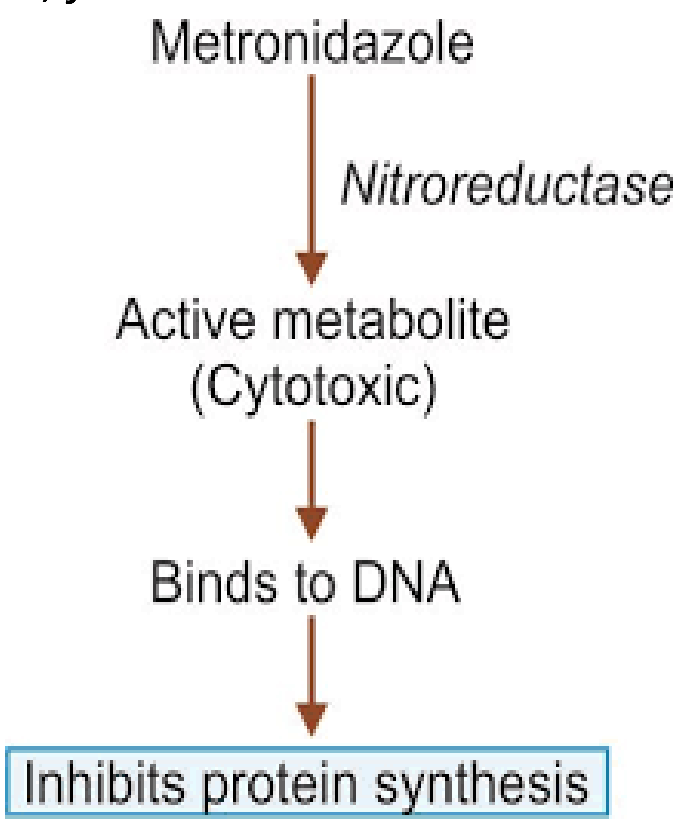
Spectrum of activity:

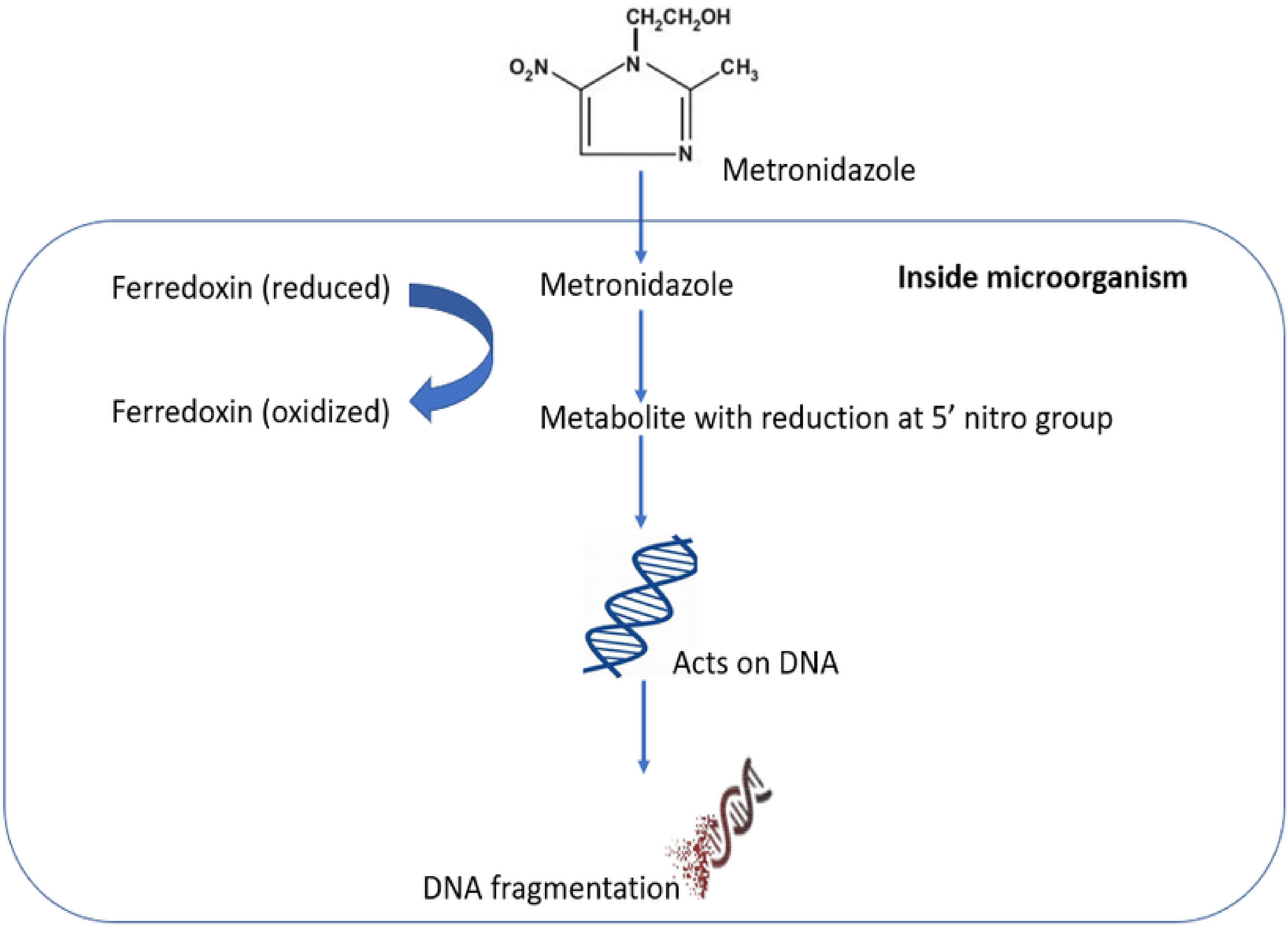
1-protozoa: *E. histolytica, giardia lamblia, trichomonas vaginalis*

2-anaerobic bacteria: *Cl difficile, Cl perfringes, H.pylori, fusobacterium bacteroides fragilis and gardenerella vaginalis*

Mechanism of Action:

Metronidazole is a prodrug requiring reductive activation of the nitro group by susceptible organisms. These organisms contain electron transport components that donate electrons to metronidazole. This reactive reduction products target the DNA and responsible for its antimicrobial activity. The mechanism of tinidazole is the same.




CH2CH2OH
O2N
N
CH3
N
Metronidazole
Inside microorganism
Ferredoxin (reduced)
Ferredoxin (oxidized)
Metronidazole
Metabolite with reduction at 5’ nitro group
Acts on DNA
DNA fragmentation

Pharmacokinetics:

Oral metronidazole and tinidazole are readily absorbed. Peak plasma concentrations are reached in 1–3 hours. Protein binding of both drugs is low (10–20%); the half-life of unchanged drug is about 8 hours for metronidazole and 12–14 hours for tinidazole.

Metronidazole distributes well throughout body tissues and fluids (vaginal and seminal fluids, saliva, breast milk, and CSF).

Metabolism depends on hepatic oxidation (CYP 450) followed by glucuronidation.

Metronidazole and its metabolites are excreted mainly in the urine. Plasma clearance of metronidazole is decreased in patients with impaired liver function.

Adverse Effects:

- Nausea, vomiting, headache, dry mouth, or a metallic taste in the mouth occurs commonly. Other side effects, rash (oral moniliasis), dysuria, dark urine, rarely, neurotoxicity (dizziness, vertigo and paresthesia).
- If taken with alcohol, **a disulfiram-like** reaction may occur, (nausea, vomiting, sweating, flushing, tachycardia, hypotension)
- Tinidazole although, has similar adverse-effect, but better tolerated.

Drug interaction:

- Phenobarbitone, rifampin (Enzyme inducers) may reduce its therapeutic effect.
- Cimetidine (Enzyme inhibitor) can reduce metronidazole metabolism its dose may need to be decreased.
- Metronidazole enhances warfarin action by inhibiting its metabolism.
- It can decrease renal elimination of lithium and precipitate toxicity

Clinical indication:

- Amebiasis , Giardiasis and Trichomoniasis
- Cutaneous leishmaniasis
- Sepsis, peritonitis, meningitis, brain abscess caused by sensitive microbes (used IV)
- Respiratory infections, infections of the intestine and urinary pathways caused by sensitive microbes
- Pseudomembranous colitis
- Prophylaxis of anaerobic infection before abdominal surgery
- Peptic ulcer associated with Helicobacter pylori
- Infections of the skin (perioral dermatitis) and soft tissues (used topically)
- Paradontitis, ulcerative stomatitis (topically).

Iodoquinol: is a halogenated hydroxyquinoline. It is effective against the luminal trophozoite and cyst forms, but not against trophozoites in the intestinal wall or extraintestinal tissues. commonly used with metronidazole to treat amebic infections

Paromomycin: is an aminoglycoside antibiotic, Only effective against the intestinal (luminal) forms of E. histolytica, because it is not significantly absorbed from the gastrointestinal tract. It is also an alternative agent for cryptosporidiosis and giardiasis.

Diloxanide furoate: It is an effective luminal amebicide but is not active against tissue trophozoites. is the drug of choice for asymptomatic luminal infections.

Chloroquine: is used in combination with metronidazole to treat amebic liver abscesses. It eliminates trophozoites in liver abscesses, but it is not useful in treating luminal amebiasis

Emetine: an alkaloid derived from ipecac, and **dehydroemetine,**

a synthetic analog, are effective against tissue trophozoites of *E histolytica*, but because of major toxicity concerns their use is limited

Nitazoxanide: was recently approved in the USA. Used for treatment of cryptosporidiosis, giardiasis, amebic and also has activity against intestinal helminthes. It inhibits the enzyme essential to anaerobic energy metabolism in protozoan and bacterial species.

Giardiasis: is caused by the flagellated protozoan *Giardia intestinalis.* Infection results from ingestion of the cyst form of the parasite, which is found in fecally contaminated water or food.

Treatment with a 5-7 day course of metronidazole usually is successful, although sometimes therapy may have to be repeated or prolonged.

A single dose of tinidazole may be superior to metronidazole. Paromomycin can be used to treat pregnant women to avoid any possible mutagenic effects of the other drugs.

Cryptosporidiosis: Cryptosporidia are coccidian protozoan parasites that cause diarrhea. *Cryptosporidium parvum* and the newly named *Cryptosporidium hominis* . Infectious oocysts in feces may be spread either by direct human-to-human contact or by contaminated water supplies.

In most individuals, infection is self-limited. However, in patients with AIDS and other immunocompromised individuals, the severity of diarrhea may require hospitalization.

Nitazoxanide has shown activity in treating cryptosporidiosis in immunocompetent children and adults

Intestinal helmenthic infestations

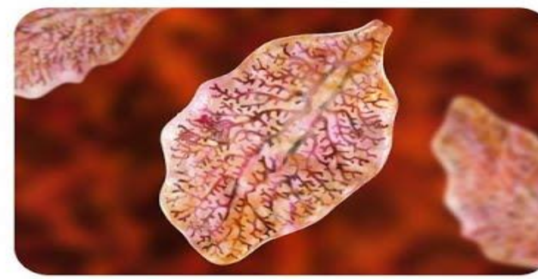
Helminths (worms): are multicellular organisms that infect very large numbers of humans and cause a broad range of diseases.

Three major groups of helminths infect humans are:

Nematodes (Roundworms), Trematodes (Flukes), and **cestodes (Tapeworms)**

Roundworm

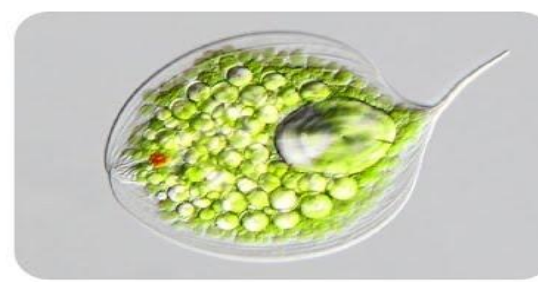
Hookworm



Fluke

Tapeworm

Nematodes

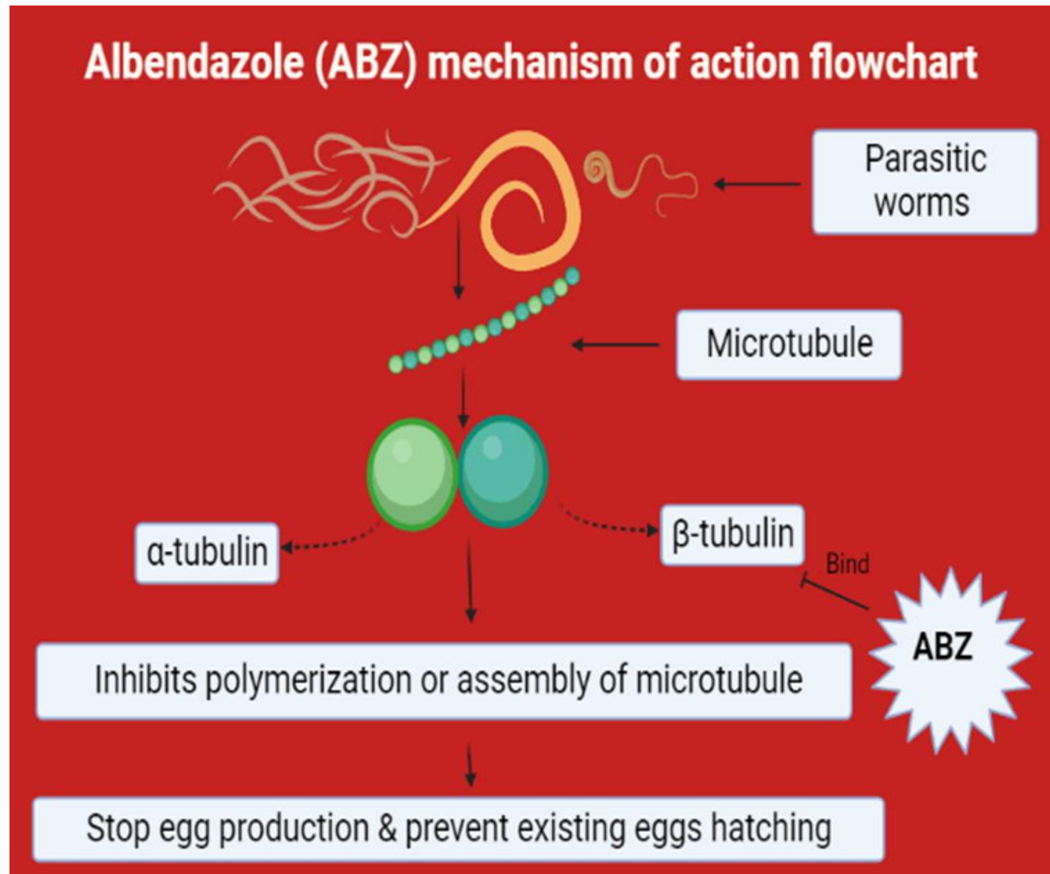


Protozoa

Drugs for treatment of Nematodes(Round worms)

Benzimidazoles(BZs): Although a large number of drugs in this class have been synthesized, only two are currently in wide use, namely, albendazole and mebendazole

Mechanism of action: It acts by inhibiting microtubule synthesis. It binds with parasite's "B-tubulin" and inhibit Its polymerization. In addition It irreversibly blocks glucose uptake by the parasite then affected parasites are expelled in the feces.



Its a first-line agent for the treatment of infections caused by whipworms (Trichuris trichiura), pinworms (Enterobius vermicularis), hookworms (Necator americanus and Ancylostoma duodenale), and roundworms (Ascaris lumbricoides).

Adverse effects include abdominal pain and diarrhea. It should be avoided in pregnancy

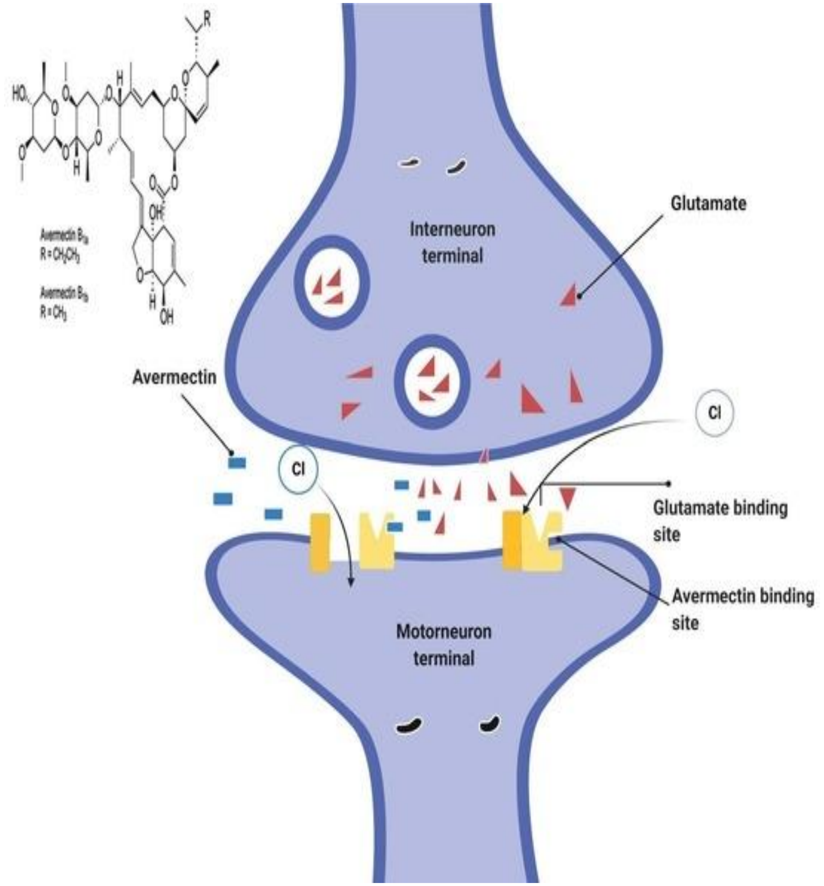
Pyrantel pamoate: is also effective in the treatment of infections caused by roundworms, pinworms, and hookworms.

Mechanism of action: It is poorly absorbed orally and exerts its effects in the intestinal tract. It acts as a depolarizing, neuromuscular-blocking agent, causing release of acetylcholine and inhibition of cholinesterase, leading to paralysis of the worm. The paralyzed worm releases its hold on the intestinal tract and is expelled.

Ivermectin: is the drug of choice for the treatment of cutaneous larva migrans, strongyloidiasis, and onchocerciasis (river blindness) caused by Onchocerca volvulus (kills microfilariae but has no activity against adult worms).

Its more effective in ascariasis and enterobiasis than in trichuriasis or hookworm infection.

It targets the glutamate-activated Cl− channels found in nematode nerve or muscle cells and causes hyperpolarization by increasing intracellular chloride concentration, resulting in paralysis and death of the worm. The drug is given orally and does not readily cross the blood–brain barrier.



Ivermectin should not be used in pregnancy.

Drugs for treatment of Trematode (Flukes)

Trematodes (flukes): are leaf-shaped flatworms. Flukes can be divided into different groups depending on where they live in the affected host as, blood flukes, liver flukes, lung flukes, or intestinal flukes.

Schistosoma, Fasciola, Clonorchis, Opisthorchis, Paragonimus species are among the trematodes that can infect humans

Praziquantel: is effective in the treatment of schistosome infections of all species and most other trematode and cestode infections.

Praziquantel appears to increase the permeability of trematode and cestode cell membranes to calcium, resulting in paralysis, dislodgement, and death.

Pharmacokinetics:. It is rapidly absorbed, with a bioavailability of about 80% after oral administration. It should be taken with food and not chewed due to a bitter taste.

Peak serum concentrations are reached 1–3 hours after a therapeutic dose, then distributes throughout the body, with highest concentrations measured in the liver and kidneys and distributes into the cerebrospinal fluid (CSF). About 80% of the drug is bound to plasma proteins.

Common **adverse effects** include dizziness, malaise, and headache as well as gastrointestinal upset.

The bioavailability of praziquantel is reduced by inducers of hepatic CYPs, such as phenytoin, carbamazepine and phenobarbital; or with corticosteroids while its plasma concentrations increase with cimetidine

Praziquantel is **contraindicated** for the treatment of ocular cysticercosis, because destruction of the organism in the eye may cause irreversible damage.

Drugs for treatment of Cestodes (Tapeworms)

Cestodes (Tapeworms): typically have a flat, segmented body and attach to the host’s intestine. The four main intestinal cestode pathogens of humans are:

- Taenia saginata (beef tape worm)
- Taenia solium (pork tapeworm)
- Hymenolepis nana (dwarf tapeworm)
- Diphyllobothrium latum (fish tapeworm)

Albendazole: inhibits microtubule synthesis and glucose uptake in nematodes and is effective against most nematodes.

Its primary therapeutic application, however, is in the treatment of cestodal infestations, such as cysticercosis and hydatid disease is also very effective in treating microsporidiosis, a fungal infection

Niclosamide: is an alternative to praziquantel for the treatment of cestode (tapeworm) infections (taeniasis, diphyllobothriasis, and others)

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